



Estimation of Sunflower Seed Yield Using Partial Least Squares Regression and Artificial Neural Network Models

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ABSTRACT

Statistical models can efficiently establish the relationships between crop growth and environmental conditions while explicitly quantifying uncertainties. This study aimed to test the efficiency of statistical models established using partial least squares regression (PLSR) and artificial neural network (ANN) in predicting seed yields of sunflower (*Helianthus annuus*). Two-year field trial data on sunflower growth under different salinity levels and nitrogen (N) application rates in the Yichang Experimental Station in Hetao Irrigation District, Inner Mongolia, China, were used to calibrate and validate the statistical models. The variable importance in projection score was calculated in order to select the sensitive crop indices for seed yield prediction. We found that when the most sensitive indices were used as inputs for seed yield estimation, the PLSR could attain a comparable accuracy (root mean square error (RMSE) = 0.93 t ha⁻¹, coefficient of determination (R^2) = 0.69) to that when using all measured indices (RMSE = 0.81 t ha⁻¹, R^2 = 0.77). The ANN model outperformed the PLSR for yield prediction with different combinations of inputs of both microplots and field data. The results indicated that sunflower seed yield could be reasonably estimated by using a small number of crop characteristic indices under complex environmental conditions and management options (e.g., saline soils and N application). Since leaf area index and plant height were found to be the most sensitive crop indices for sunflower seed yield prediction, remotely sensed data and the ANN model may be joined for regional crop yield simulation.

Key Words: leaf area index, microplot experiment, plant height, remote sensing, salinization, variable importance in projection score

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INTRODUCTION

As a widely planted crop for the production of food and oil, more than 25 million hectares of sunflower (*Helianthus annuus*) are distributed across the world croplands (National Sunflower Association, 2013). Since it is well adapted to moderately saline soil conditions (Francois, 1996), sunflower is one of the best crops for arid or semiarid regions undergoing soil salinization (Zeng *et al.*, 2014). The Hetao Irrigation District, Inner Mongolia, China, is one of the regions suffering from salinity due to low precipitation and high evapotranspiration. Approximately 70% of the cultivated lands there are suffering from salinization (Shi *et al.*, 2012; Li *et al.*, 2014). The profit from selling sunflowers is becoming a major source of income for local farmers. In 2009, according to the local government, sunflower has grown on approximately 0.18 million hectares in Hetao Irrigation District, with a yield generated more than 1.5 billion yuan (Chinese RMB) in revenue. Esti-

ating sunflower seed yield in saline soils is essential to perform technico-economic analysis to support small-hold farmers in practicing decisions.

Farm survey is an efficient way to identify farm management practices, soil conditions, and crop performance, but may fail to capture their interactions at a regional scale due to the high spatial heterogeneity of soil conditions and the wide range of field management practices performed by large numbers of farmers (Morel *et al.*, 2014). Consequently, alternative techniques for seed yield estimation must be developed. A process-based crop growth model is an alternative method because of its ability in simulating the interactions between crop, soil, atmosphere, and management practices (Easterling *et al.*, 1996; Sinclair and Seligman, 1996; Jones *et al.*, 2003; Zhao *et al.*, 2013). For example, many of these models have been applied to evaluate crop yield under different production conditions and optimize nitrogen (N) and water management according to local climate-soil conditions (Diepen *et al.*,

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1989; Wolf *et al.*, 2011; de Wit *et al.*, 2012; Boogaard *et al.*, 2013). However, these models require a large number of detailed crop and soil parameters to reflect the processes in agricultural systems. For large-scale applications, many of these parameters are not available or difficult to be derived, resulting in considerable uncertainties in simulations (Lobell, 2013).

Remote sensing is becoming an efficient way to collect data of high spatial-temporal resolution, *e.g.*, normalized difference vegetation index (NDVI) for crop development (Idso *et al.*, 1977; Glenn *et al.*, 2011). Because the assimilation of remotely sensed data (*e.g.*, soil moisture, NDVI, and leaf area index (LAI)) to process-based models is difficult and has considerable computing demands, statistical models may be more suitable in estimating crop yields at a large scale (Mkhabela *et al.*, 2005; Xiao *et al.*, 2005; Mello *et al.*, 2014). For example, NDVI has been used to estimate yields for rice, maize, and sugarcane (Mello *et al.*, 2014). Biophysical canopy variables such as LAI can be derived from remotely sensed data and linked with yield estimation (Gitelson *et al.*, 2014).

Partial least squares regression (PLSR) and artificial neural networks (ANN) are two frequently used methods to construct statistical crop models and to establish a link between environmental conditions and crop growth (Farifteh, 2007; Farifteh *et al.*, 2007; Pahlavan *et al.*, 2012; Abdel-Rahman *et al.*, 2014). The PLSR was developed for econometric modeling of multivariate time-series data and was later applied to chemometrics (Wold *et al.*, 2001) and hyper-spectral remote sensing (Cho *et al.*, 2007). It can extract relevant parts of large data matrices to solve the problem of collinearity and produce more reliable models compared to other calibration methods (Thomas and Haaland, 1990). Recently, PLSR has been used to predict plant biomass, LAI, and crop density using reflectance measurements (Hansen and Schjoerring, 2003). In addition, the variable importance in projection (VIP) score calculated by PLSR can provide a useful indication of parameter sensitivity in response to desirable outputs (Oussama *et al.*, 2012). Similar to PLSR, ANNs are designed for dynamic nonlinear systems (Dai *et al.*, 2011). It is robust to recurrent changes and can detect systematic changes in complex natural systems. It has been increasingly used in crop research including crop yield prediction (Kaul *et al.*, 2005), evapotranspiration estimation (Kumar *et al.*, 2002), and soil salinity impact assessments (Zou *et al.*, 2010). However, to the best of our knowledge, no study has explicitly compared the ability of aforementioned two methods, PLSR and ANN, in constructing statistical models for

estimating crop production.

Accordingly, this study aimed to compare the efficiency of PLSR and ANN in establishing statistical crop models for predicting crop yields under various environmental conditions and N application rates. We firstly selected the most sensitive crop indices for estimating sunflower seed yields. The statistical models linking the sensitive crop indices and yields were established using two years of microplot experimental data. The performances of the two models were evaluated and compared using the aforementioned microplot experimental data as well as field trial data.

MATERIALS AND METHODS

Field experimental design

A field experiment was conducted at the Yichang Experimental Station in Hetao Irrigation District (40°19'–41°18' N, 106°20'–109°19' E), Inner Mongolia, China (Fig. 1a). Temperate continental monsoon dominates the climate in this region. The mean annual precipitation varies from 139 to 222 mm, with approximately 60% falling in July and August. The mean annual potential evapotranspiration is nearly 10 times higher than the precipitation, varying from 2 200 to 2 400 mm (Li *et al.*, 2010). The strong evaporative forces transport groundwater upward, resulting in salt accumulation across soil surface layers (Feng *et al.*, 2005). Two years of experiments from 2012 to 2013 were conducted on sunflowers. The crop was planted in 36 microplots with undisturbed soil (Fig. 1b). Each microplot covered an area of 1.8 m × 1.8 m, and sides were wrapped with impermeable plastic to a depth of 1.5 m, allowing for free drainage. Electrical conductivity (EC, dS m⁻¹), a key soil salinity indicator, in the soil top layers (0–20 cm) was measured in each block before sowing at a soil:water ratio of 1:5 (weight:volume) (EC_{1:5}) using a digital conductivity meter. The EC_{1:5} was converted into EC of the saturated paste extract (EC_e) using the equation: $EC_e = 7.4 \times EC_{1:5}$ (Hachicha *et al.*, 2005). According to the FAO standard for soil salinity classification (FAO, 1985), the soil salinity level was designated as non-salinized (S₀, EC_e = 2.44–4.44 dS m⁻¹), low (S₁, EC_e = 4.45–9.03 dS m⁻¹), moderate (S₂, EC_e = 9.04–18.07 dS m⁻¹), and high (S₃, EC_e = 18.08–29.23 dS m⁻¹) in 2012 and as non-salinized (S₀, EC_e = 2.32–4.44 dS m⁻¹), low (S₁, EC_e = 4.45–8.93 dS m⁻¹), and moderate (S₂, EC_e = 9.11–11.85 dS m⁻¹) in 2013. Three N rates were applied in 2012, including low (90 kg ha⁻¹), moderate (135 kg ha⁻¹), and high (180 kg ha⁻¹). Four N rates including 0, 90, 135, and 180 kg ha⁻¹ were applied in 2013. Nitrogen was app-

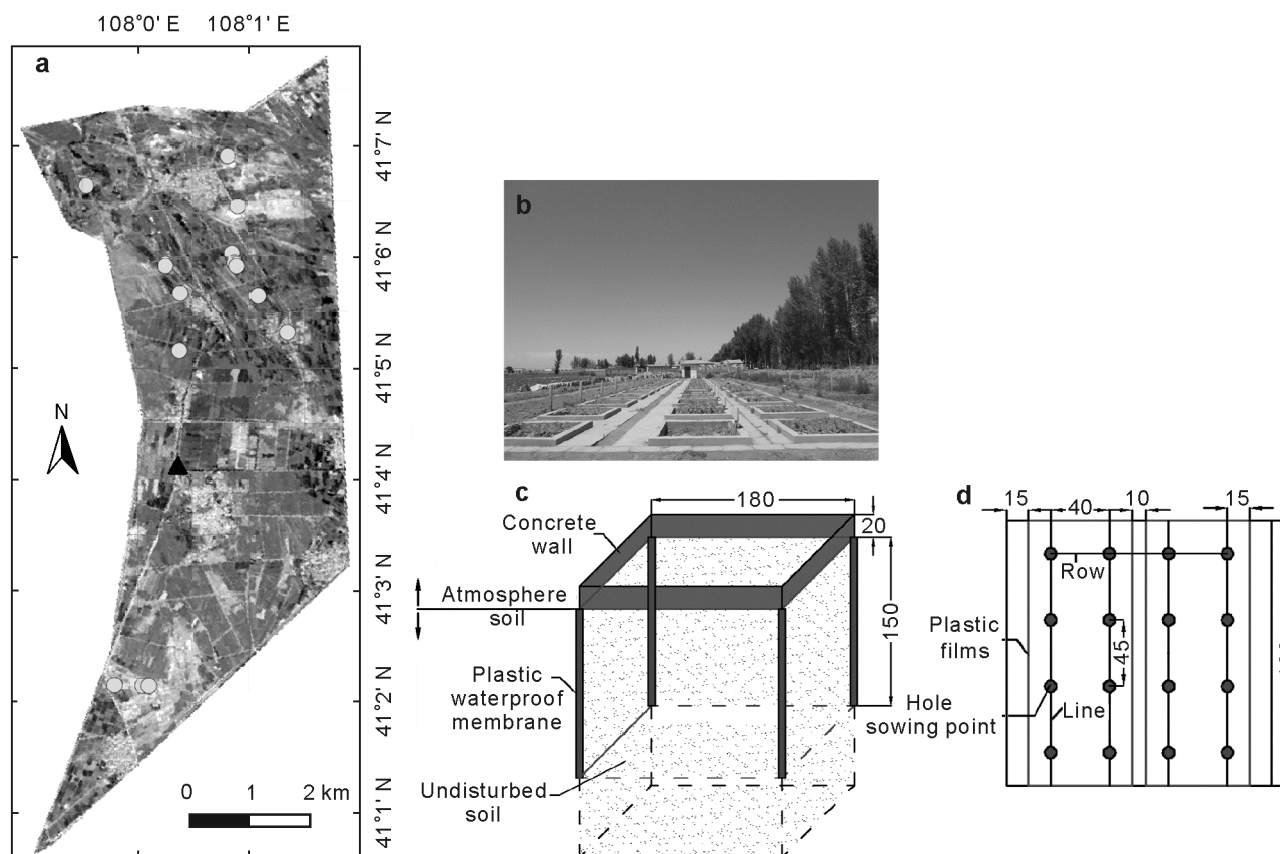


Fig. 1 Description of the microplot experiment at the Yichang Experimental Station in Hetao Irrigation District, Inner Mongolia, China: location of the study area (triangle represents the location of the microplot experiment, and circles represent the 14 fields around) (a), arrangement of the trial blocks (b), schematic diagram of the block profile (unit: cm) (c), and plant arrangement in each block (unit: cm) (d).

lied one day before sowing during each of the two years. These treatments were assigned to microplots using a randomized complete block design with three replicates (Fig. 1c). Sunflower (cv. LD5009) was planted on June 7, 2012 and June 4, 2013 in a grid of four rows ($0.45 \text{ m} \times 0.4 \text{ m}$ spacing) in each microplot (Fig. 1d), where the plant density was $4.94 \text{ plants m}^{-2}$. Plants were harvested on September 24, 2012 and September 16, 2013. Weed pest and disease were well controlled for both years. The climate conditions during the sunflower growth period are shown in Table I.

Measurement of sunflower crop characteristics

In 2012 and 2013, characteristics, including sunflower seed yield (SY), LAI, plant height (H), and soil and plant analyzer development (SPAD) value, were measured during four growth stages (seedling, bud, flowering, and mature). Bud or flower diameter (D) was measured at budding, flowering, and mature stages, and total aboveground dry matter (TADM) was measured at harvest. Table II shows the details of the measurements. The length (L) and width (W) of each leaf were determined using a measuring tape, and the leaf

TABLE I

Climate conditions^{a)} at the Yichang Experimental Station in Hetao Irrigation District, Inner Mongolia, China during sunflower growth periods in 2012 and 2013

Year	Month	T_{mean}	T_{max}	T_{min}	RH	WS	SR	Precipitation
		$^{\circ}\text{C}$			%	m s^{-1}	W m^{-2}	mm
2012	Jun.	6.7	34.4	7.7	58.1	1.8	276.3	88.0
	Jul.	6.1	33.9	11.9	73.1	1.2	213.8	99.6
	Aug.	7.0	32.8	5.4	70.0	1.2	232.8	36.8
	Sep.	7.0	27.0	-1.2	70.6	1.3	171.0	42.0
	Oct.	7.8	23.9	-7.4	60.8	1.6	143.7	13.4
2013	Jun.	7.1	35.6	8.4	58.2	1.8	220.0	25.4
	Jul.	7.1	35.2	11.7	69.0	1.2	222.5	8.8
	Aug.	7.7	35.2	5.0	69.6	1.3	215.9	27.6
	Sep.	8.7	30.2	-1.9	63.8	1.5	182.0	24.4
	Oct.	8.5	28.8	-8.5	59.1	1.4	152.6	2.6

^{a)} T_{mean} , T_{max} , and T_{min} = mean, maximum, and minimum temperature, respectively; RH = relative humidity; WS = wind speed; SR = solar radiation.

area was calculated as follows: leaf area = $0.6564 \times L \times W$ (Chen, 1984). The LAI was defined as the leaf area per unit ground surface area. Sunflower SY was calculated based on the air-dried sunflower seed with

a moisture content $< 80 \text{ g kg}^{-1}$. The SPAD value was measured using a SPAD-502 meter (Ministry of Agriculture, Japan); H and D were determined using a measuring tape; TADM was measured by drying plants at 70°C to a constant weight, followed by weighing.

TABLE II

Time for measurements of sunflower characteristics for the microplot experiment conducted at the Yichang Experimental Station in Hetao Irrigation District, Inner Mongolia, China in 2012 and 2013

Crop characteristic ^{a)}	Days after sowing ^{b)}	
	2012	2013
SY	109	104
LAI	45, 56, 63, 74	36, 50, 71, 95
SPAD value	45, 56, 63, 74	36, 50, 71, 95
H	45, 56, 63, 74	36, 50, 71, 95
D	56, 63, 74	50, 71, 95
TADM	109	104

^{a)}SY = seed yield; LAI = leaf area index; SPAD = soil and plant analyzer development; H = plant height; D = diameter of bud or flower; TADM = total aboveground dry matter.

^{b)}SY and TADM were measured at harvest; LAI, SPAD value, and H were measured at the seedling, bud, flowering, and mature stages; D was measured at budding, flowering, and mature stages.

In addition, we selected 14 fields around the Yichang Experimental Station that also planted sunflowers (cv. LD5009) in 2013. We measured H at each growth stage and SY at harvest in each field. We also randomly selected 30 sunflowers in each field, and the mean H and SY of these 30 plants were regarded as the typical values for this field. Details of the 14 fields are provided in Table III.

Partial least squares regression model

One aim of using PLSR in this study was to select sensitive variables for SY estimation. Therefore, unlike the common approaches that divide datasets into two groups, we used all data for model establishment. Sunflower SY was treated as a dependent variable (y) and other crop characteristic indices as independent variables (x). The PLSR is based on the principle of variable compression, which strives to identify a few linear combinations to generate new components of the original x value that describe most of the inherent variation in independent variables. Mathematically, this is achieved by maximizing the covariance between y and all possible linear functions of x . Therefore, a leave-one-out cross-validation method was used to determine the appropriate component numbers. In addition, root mean square error (RMSE) in the cross-validation process (RMSECV, Eq. 1) and the explained variances of

TABLE III

Sunflower seed yield (SY) and plant height (H) in 14 fields around the Yichang Experimental Station in Hetao Irrigation District, Inner Mongolia, China in 2013

Field number	Longitude	Latitude	H			SY
			Bud	Flowering	Mature	
	°		cm			t ha ⁻¹
1	108.006 13	41.085 98	147	180	183	5.48
2	108.006 24	41.094 58	157	144	165	6.27
3	108.004 04	41.098 71	137	151	181	5.92
4	107.992 15	41.110 74	100	110	115	4.62
5	108.013 42	41.115 18	127	134	150	5.43
6	108.014 90	41.107 64	165	150	152	7.23
7	108.014 07	41.100 59	118	137	123	4.35
8	108.014 47	41.099 16	146	160	159	7.41
9	108.014 74	41.098 70	142	150	127	5.53
10	108.018 01	41.094 22	120	100	123	4.32
11	108.022 40	41.088 76	135	142	150	5.63
12	107.996 41	41.035 83	92	134	137	4.91
13	108.000 60	41.035 77	118	105	130	5.12
14	108.001 54	41.035 72	118	114	129	4.95

x and y values (hereafter referred to as EVX and EVY, respectively) were calculated.

$$\text{RMSECV} = \sqrt{\frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n}} \quad (1)$$

where $\hat{y}$ indicates the values of the y variable that are estimated by cross-validation, and the $\hat{y}_i$ value for object i is estimated using a model that was built using a set of objects excluding object i ; y_i indicates the known values of the y variable for object i ; and n is the total number of objects in the data set.

To determine the major characteristics for yield estimation, the VIP score was calculated for each variable as follows:

$$\text{VIP}_k(a) = K \sum_a w_{ak}^2 \left(\frac{\text{SSY}_a}{\text{SST}_t} \right) \quad (2)$$

where $\text{VIP}_k(a)$ is the VIP score of the k th predictor variable based on a model with factor a , w_{ak} is the corresponding loading weight of the k th variable in the a th PLSR factor, SSY_a is the explained sum of squared y by a PLSR model with factor a , SST_t is the total sum of squared y , and K is the total number of predictor variables. The crop characteristic indices (x variables) with a VIP score greater than 1.0 for all components are considered important (Chong and Jun, 2005). In this study, the PLS library (Meivik and Wehrens, 2007) in the R statistical package version 2.15.2 (R Development Core Team, 2011) was used to implement the

PLSR algorithm.

Artificial neural networks

A three-layer (*i.e.*, input, hidden, and output) ANN was used in this study (Fig. 2). The input layer contained the most sensitive variables to yields, which were identified by the PLSR method. The output layer was the output variable for crop yields. The hidden layer included functions used to map the input variables to the output variable. We chose the backpropagation (BP) learning algorithm to train the three-layer ANN (Traore *et al.*, 2010; Dai *et al.*, 2011). Progressing forward from the hidden layer to the output layer and then backward from the input layer to the hidden layer, the BP learning algorithm first calculated the gradients of the cost function with all current weights and then returned to calculate and update the weights according to the gradients of the cost function. In the forward calculation, each input was weighted before being fed into the hidden layer. We used the general sigmoidal logistic transfer function defined as $1/(1 + e^{-x})$, which can express many complex relationships (Smith, 1993). The computation of the output was conducted layer by layer from the input to the output in a forward direction, and then the error between the desired and actual output was computed. In the BP pass, the interconnecting weights were adjusted during training using BP of errors, in which the weight associated with each connection was adjusted by an amount proportional to the strength of the signal in the connection and the total measure of the error. A detailed description of the BP learning algorithm can be found in Hecht-Nielsen (1989).

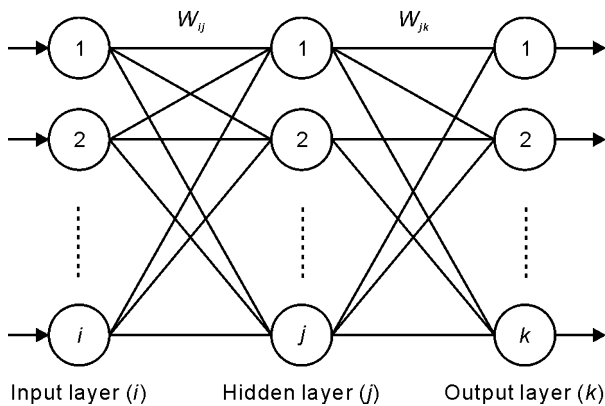


Fig. 2 Three-layer feed-forward artificial neural network architecture. W_{ij} and W_{jk} indicate the weights between the neurons in the adjacent layers.

We used bootstrap resampling to divide the data into two subsets for model training and testing (Hall and Martin, 1988). We bootstrapped the processes 100

times and calculated the average simulated SY as the final result to improve the accuracy of the BP neural network simulation. The size of hidden nodes was another factor that affected the simulation accuracy. However, the greater the size of the hidden nodes, the more time was required for a single computation. Therefore, we found that a size of six could achieve good simulation accuracy with an acceptable computing time.

Step of model establishment and validation

The experimental data from microplots were used for sensitive index selection and model establishment. In selecting indices sensitive to SY, we regarded SY as a dependent variable and all other measure crop indices as independent variables to estimate SY using PLSR (hereafter referred to as PLSR1). The VIP scores for each component were then calculated. The crop indices whose minimum VIP scores for all components were greater than 1.0 were marked as sensitive. Selected sensitive crop indices were used to establish an SY estimation model using PLSR (hereafter referred to as PLSR2). In addition, similar to the PLSR models, two ANN models (hereafter referred to as ANN1 and ANN2) were also established for SY estimation with the same variables according to the two PLSR models. After this step, the field data were used to test the performances of the PLSR2 and ANN2 models.

Model evaluation criteria

The RMSE (Eq. 3) and coefficient of determination (R^2) (Eq. 4) were used to evaluate model efficiency as follows:

$$\text{RMSE} = \sqrt{\sum_{i=1}^n \frac{(y_i^{\text{obs}} - y_i^{\text{sim}})^2}{n}} \quad (3)$$

$$R^2 = \left(\frac{\sum_{i=1}^n (y_i^{\text{obs}} - \bar{y}_i^{\text{obs}})(y_i^{\text{sim}} - \bar{y}_i^{\text{sim}})}{\sqrt{\sum_{i=1}^n (y_i^{\text{obs}} - \bar{y}_i^{\text{obs}})^2} \sqrt{\sum_{i=1}^n (y_i^{\text{sim}} - \bar{y}_i^{\text{sim}})^2}} \right)^2 \quad (4)$$

where y_i^{obs} is the i th observed SY, y_i^{sim} is the i th simulated SY, and $\bar{y}_i^{\text{obs}}$ and $\bar{y}_i^{\text{sim}}$ are the mean of y_i^{obs} and y_i^{sim} , respectively.

RESULTS

Sensitive crop index selection and SY prediction

When all observations from microplots of expe-

perimental data were used as independent variables in the PLSR analysis, the smallest RMSECV (1.01 t ha^{-1}) was approached by 12 components in the cross-validation process, and both EVX and EVY were greater than 77% (Fig. 3a). Equation 5 represents the PLSR using all crop indices as independent variables (PLSR1):

$$\begin{aligned} \text{SY} = & 2.526\text{SPADS} - 0.522\text{SPADB} + 0.517\text{SADF} + \\ & 0.069\text{SPADM} + 0.350\text{LAIS} + 3.340\text{LAIB} + \\ & 7.726\text{LAIF} + 10.006\text{LAIM} - 0.076\text{HS} - \\ & 0.033\text{HB} - 0.889\text{HF} + 0.0004\text{HM} + \\ & 0.953\text{DB} + 0.484\text{DF} + 0.493\text{DM} + \\ & 0.005\text{TADB} - 172.279 \end{aligned} \quad (5)$$

where SPADS, SPADB, SAPDF, and SPADM indicate the SPAD values at seedling, bud, flowering, and

mature stages, respectively; LAIS, LAIB, LAIF, and LAIM indicate LAI at seedling, bud, flowering, and mature stages, respectively; HS, HB, HF, and HM indicate H at seedling, bud, flowering, and mature stages, respectively; and DB, DF, and DM indicate the D of bud or flower at bud, flowering, and mature stages, respectively. Equation 5 simulated the SY with high accuracy as compared to observations in the microplot experiment ($\text{RMSE} = 0.81 \text{ t ha}^{-1}$, $R^2 = 0.77$, Fig. 3b).

The VIP scores for each component in the microplot experiment were calculated after PLSR analysis in order to select sensitive crop characteristic indices (minimum VIP score > 1). The H and LAI at bud, flowering, and mature stages were found to be sensitive to sunflower SY prediction (Fig. 4), and the H and LAI at the mature stage were the two most sensitive indices.

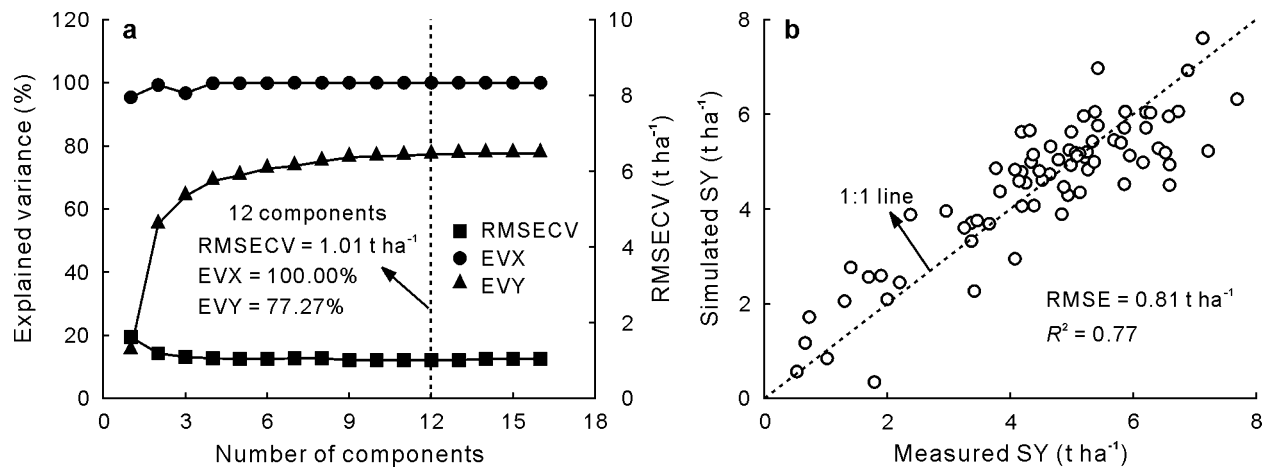


Fig. 3 Performance of the partial least squares regression model in estimating the sunflower seed yield (SY) with different components (a) and all measured sunflower characteristics (b). RMSECV = root mean square error in the cross-validation process; EVX and EVY = explained variances of x and y values, respectively; RMSE = root mean square error between simulated and measured values; R^2 = coefficient of determination.

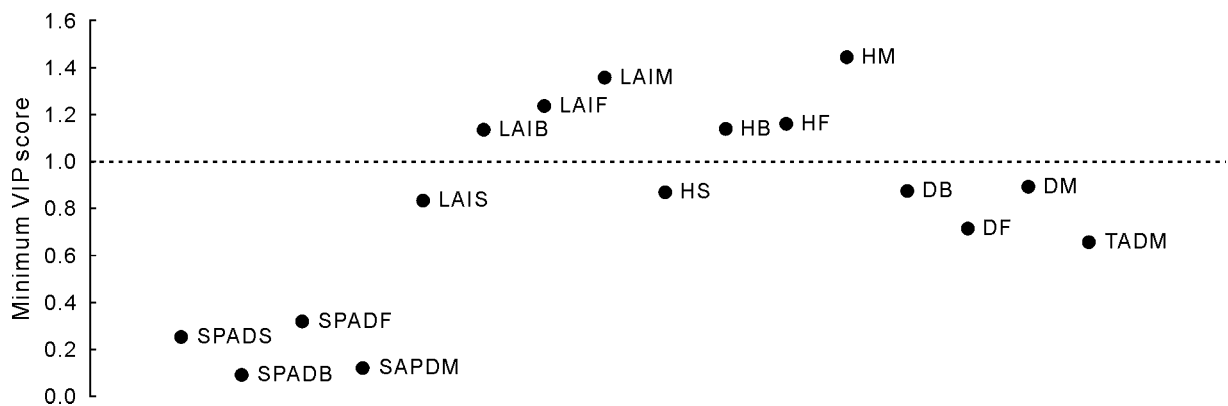


Fig. 4 Variable importance in projection (VIP) scores and selection of sensitive sunflower characteristics for seed yield prediction. SPADS, SPADB, SPADF, and SPADM = soil and plant analyzer development (SPAD) values at seedling, bud, flowering, and mature stages, respectively; LAIS, LAIB, LAIF, and LAIM = leaf area index at seedling, bud, flowering, and mature stages, respectively; HS, HB, HF, and HM = plant height at seedling, bud, flowering, and mature stages, respectively; DB, DF, and DM = diameter of bud or flower at bud, flowering, and mature stages, respectively; TADM = total aboveground dry matter.

Therefore, when using the two most sensitive indices at three development stages (*i.e.*, budding, flowering, and maturity) as independent variables in the PLSR analysis, we could establish a new PLSR model, PLSR2 (Eq. 6):

$$\text{ST} = 2.965\text{LAIB} + 3.939\text{LAIF} + 5.282\text{LAIM} + 0.217\text{HB} - 0.603\text{HF} + 0.001\text{HM} - 46.983 \quad (6)$$

The optimal component number was 4 with the lowest RMSECV (1.01 t ha^{-1}), although the EVX was nearly 100%, and the EVY was only 69.39% (Fig. 5a), which generated a relatively low accuracy of PLSR2 ($R^2 = 0.69$, $\text{RMSE} = 0.93 \text{ t ha}^{-1}$, Fig. 5b). The ANN models outperformed PLSR in terms of SY estimation. When compared to the PLSR1 model, the ANN1 model (using the same independent variables as the PLSR1 model) had a higher R^2 and lower RMSE

(Fig. 6a). Compared to the PLSR2 model, the ANN2 model (using the same independent variables as the PLSR2 model) increased R^2 by 17.39% and decreased RMSE by 19.35% (Fig. 6b).

Model validation

Using the data of microplot experiment, we found that the relationship between LAI and H can be fitted by linear regression analysis (Fig. 7), which was similar to Dong (2006). Therefore, we could firstly obtain LAI at different growth stages of 14 fields from H and then use them to validate our SY prediction models.

When using 14 field observations to validate PLSR and ANN models with selected sensitive indices (models PLSR2 and ANN2), the superiority of model ANN2 was even more pronounced. For model PLSR2, the R^2 was only 0.37 and RMSE was 1.04 t ha^{-1} (Fig. 8a). However, the R^2 and RMSE of model ANN2

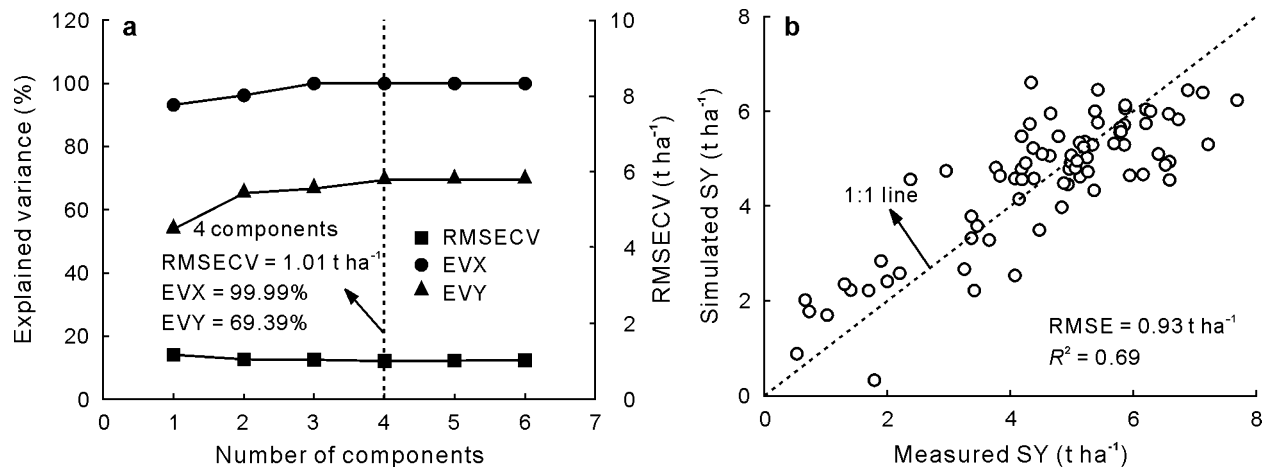


Fig. 5 Performance of the partial least squares regression model in estimating the sunflower seed yield (SY) with different components (a) and two most sensitive sunflower characteristics (b) at budding, flowering, and maturity stages. RMSECV = root mean square error in the cross-validation process; EVX and EVY = explained variances of x and y values, respectively; RMSE = root mean square error between predicted and measured values; R^2 = coefficient of determination.

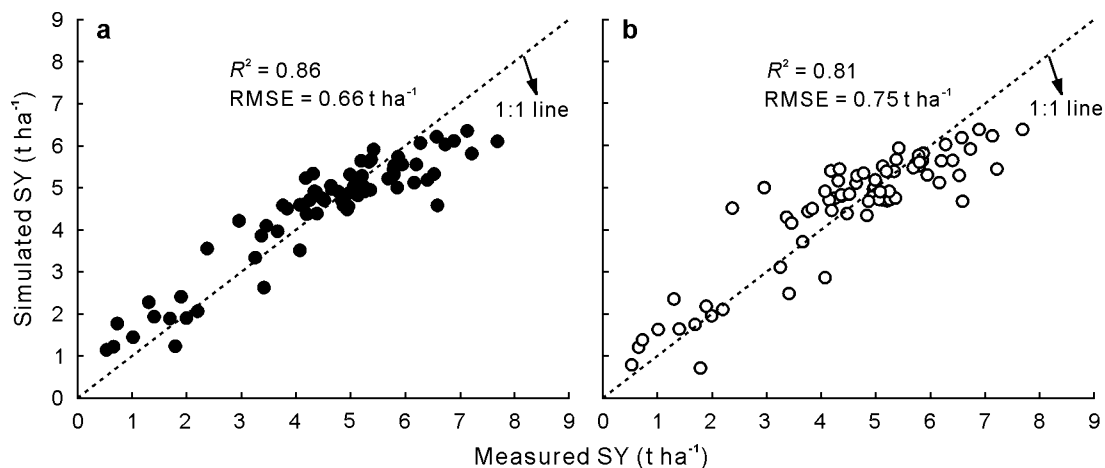


Fig. 6 Performance of the artificial neural network model in estimating the sunflower seed yield (SY) with all measured (a) or sensitive (b) sunflower characteristics. RMSE = root mean square error; R^2 = coefficient of determination.

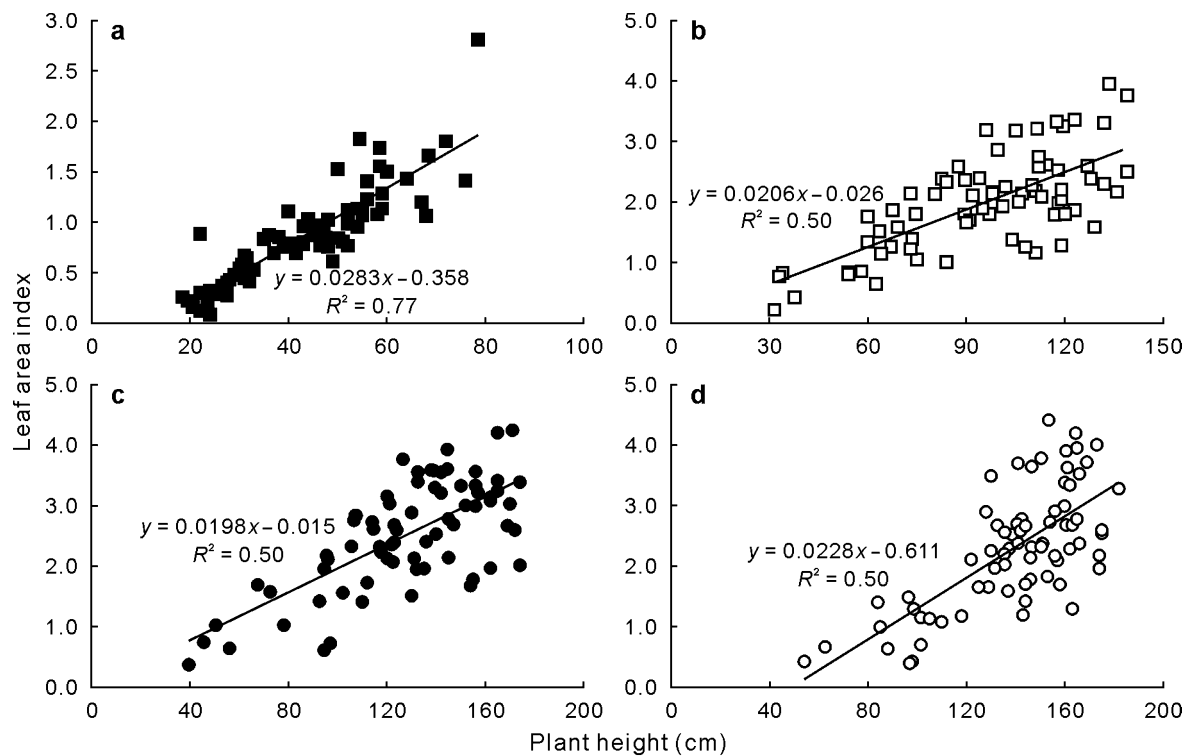


Fig. 7 Relationship between leaf area index and plant height at seedling (a), bud (b), flowering (c), and mature (d) stages of sunflower.

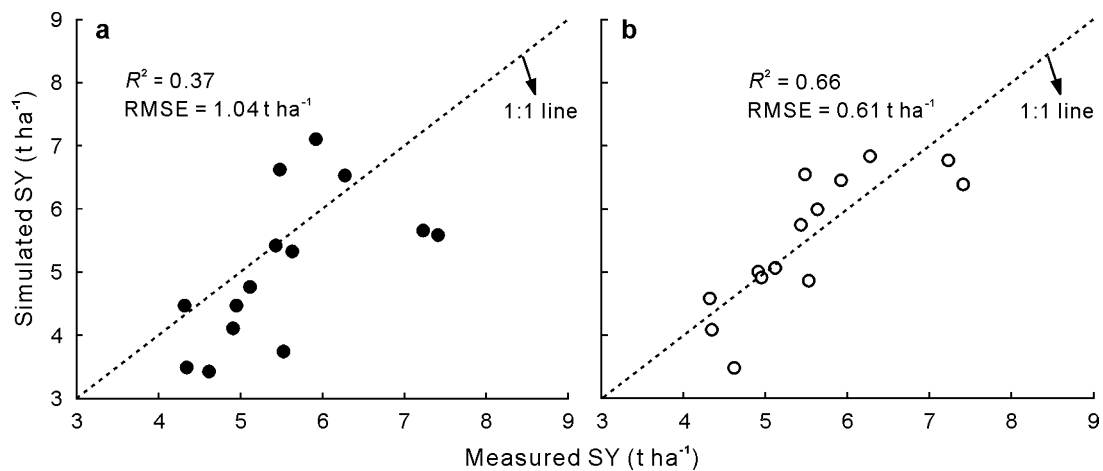


Fig. 8 Validation of the partial least squares regression model (a) and the artificial neural network model (b) in estimating the sunflower seed yield (SY) with sensitive sunflower characteristics from the 14 sunflower fields around the Yichang Experimental Station in Hetao Irrigation District, Inner Mongolia, China in 2013. RMSE = root mean square error; R^2 = coefficient of determination.

were 0.66 and 0.61 t ha⁻¹, respectively (Fig. 8b), which indicated that the ANN2 model was not only precise but also stable and could be applied to SY prediction in large scales.

DISCUSSION AND CONCLUSION

Yield estimation can provide valuable information for prescribing management practices and achieving sustainable profits under changing environments. Because crop characteristics can directly reflect crop

growth conditions and can be easily measured, they have been widely used for yield estimation in previous studies using both process-based and statistical models. For example, Clevers (1997) estimated LAI from optical remote sensing data and used it to predict sugar beet yield. Prasad *et al.* (2006) derived various indices from remote sensing data such as NDVI and a vegetation condition index, with which multivariate regression analysis was used to estimate corn and soybean yields in Iowa, USA. Statistical models can be used to model crops and processes for which process-based

models have not been developed (Lobell, 2013). They require less intensive input datasets and can easily quantify the uncertainties using techniques like bootstrapping. Statistical models can be constructed using various methods. In the present study, we compared the performances of statistical models established by PLSR and ANN.

The less the input data that are required for a crop model, the more suitable the model for large-scale applications, as data availability frequently becomes a limitation (Lobell and Ortiz-Monasterio, 2006). In this study, the number of variables was first reduced according to their importance (VIP scores) for SY estimation (Zhang and He, 2013). In the PLSR analysis, using two variables instead of 12 caused only a minimal loss in accuracy ($R^2 = 0.69$ versus 0.77). In line with this study, Fu *et al.* (2014) used four–five variables in the PLSR analysis to estimate winter wheat biomass, and Abdel-Rahman *et al.* (2014) used two variables to predict Swiss chard yield. Although fewer variables in yield simulation might lead to slightly less accuracy, the effort and cost associated with data collection could be considerably decreased. This is of paramount importance for large-scale applications, where data scarcity and uncertainty are usually the major issues. The VIP scores have proved to be a useful index that could considerably reduce input parameters in the PLSR analysis and determine the sensitive characteristics for yield estimation.

Many crop characteristics (*e.g.*, LAI) can now be obtained using remote sensing technology. Therefore, PLSR2 using only LAI and H can be integrated with remote sensing technology for large-scale applications (Gitelson *et al.*, 2014; Ma *et al.*, 2014; Tilly *et al.*, 2014). In our study, although PLSR2 was slightly less accurate than PLSR1, it was more accurate than the linear regression analysis with LAI or H . For example, when LAI or H was used as an independent variable in linear regression, the highest R^2 was only 0.51 or 0.61, respectively, whereas the R^2 of PLSR2 was 0.69. The relatively poor accuracy of PLSR2 was possibly caused by non-uniform salinity levels in 2012 and 2013 microplot experimental data. Additional two factors might also reduce prediction accuracy. Firstly, as SY in our study was not estimated at 0% moisture, but after drying (< 8% moisture), which could cause some measuring differences among treatments. Secondly, the size of the microplot might cause a border effect on the experimental units, and thus some heterogeneity on the plants growing inside them, thereby increasing the noise. These factors should be improved in future studies. We presumed that LAI and H at bud, flowering,

and maturity stages of sunflower were more sensitive to salt stress, but the soil salinity level varied more widely in 2012 ($EC_e = 2.44\text{--}29.23$ dS m $^{-1}$) than 2013 ($EC_e = 2.44\text{--}11.85$ dS m $^{-1}$). We compiled data from 2012 and 2013 to develop the model PLSR2. Therefore, LAI and H can reflect SY more accurately in salt-affected soils. However, in non-salinized soils, more crop characteristics should be considered to improve the accuracy of yield estimation.

Compared to the PLSR analysis, ANN performed better regardless of whether the number of variables was reduced, or all measured crop characteristics were used. In this study, a simple ANN model with one hidden layer and six hidden nodes could reasonably predict SY for different soil salinity levels and N application treatments. This agrees with other studies reporting that the ANN models should contain the lowest number of hidden nodes possible for simulation to avoid an oversized network (Xu *et al.*, 2011; Moosavizadeh-Mojarad and Sepaskhah, 2012). We further compared the performance of ANN with different numbers of hidden nodes and found that more than six nodes required notably greater computing cost with marginal improvement in accuracy. In contrast to other studies that divided data into training and testing groups randomly, we used bootstrap resampling to determine the training and testing groups (Hall and Martin, 1988) and used approximately 48 and 24 observations as training and testing data, respectively, in the ANN simulations. This was in agreement with Schultz *et al.* (2000) who suggested that smaller data sets lead to smaller networks and fewer weight estimates, which avoid oversized networks and possible overtraining. Similarly, Kang (1991) suggested that ANN models did not require large data sets to achieve acceptable accuracy. We performed 100 bootstrap resamplings 100 times and calculated the average value in each ANN model, which could improve simulation accuracy and make our model more stable and reliable. Using all sunflower characteristics as independent variables, the ANN1 and PLSR1 models had similar simulation accuracy. However, when the selected characteristics were used as independent variables, the ANN model performed better. Furthermore, we used all observations to establish the PLSR model and did not divide the data into calibration and validation groups. Therefore, the accuracy of these models was unrealistically high. Thus, the ANN models were more suitable for estimating SY under our experimental conditions than the PLSR models. Moreover, the validation when using data of 14 fields further indicated that model ANN2 was more accurate and stable than PLSR2. Therefore, this

study further demonstrated that SY could be precisely estimated by a small number of crop characteristics even under complex conditions (*e.g.*, saline soils and different N applications). The LAI and *H* are sensitive to yields, and these characteristics can be easily obtained by remote sensing technology (Freeman *et al.*, 2007; Tang *et al.*, 2014). Therefore, we recommend combining remote sensing technology and ANN for regional crop yield estimation and agriculture management.

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